LayerCFG: Frequency-Selective Classifier-Free Guidance for Reducing Texture Artifacts in Diffusion Models

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Abstract

Classifier-free guidance (CFG) controls class-conditional generation in diffusion models by scaling the difference $\Delta = \epsilon_c - \epsilon_u$ between conditional and unconditional noise predictions. A single scale s is applied to Δ uniformly at every denoising step, which implicitly treats all spatial frequencies of the guidance signal as equivalent. This is unlikely to be ideal because the UNet that produces Δ is hierarchical, meaning its bottleneck layers carry coarse class structure, while its decoder layers carry texture and edge information, and both are folded into Δ before the scale is applied. We ask whether the texture artifacts that appear at large s can be attributed specifically to over-amplification of the high-frequency component of Δ, independent of the semantic component. To study this, we propose **LayerCFG**, which decomposes Δ into low-frequency (semantic) and high-frequency (texture) bands via a 2-D FFT and applies independent scales s_{sem} and s_{tex} to each. We also introduce NHFP (Normalized High-Frequency Power), a metric that measures how much a set of generated images deviates from real data in terms of relative high-frequency energy, something FID does not directly capture. On CIFAR-10, a 16-point sweep over $(s_{\text{sem}},\ s_{\text{tex}})$ reveals that the two bands are largely decoupled: s_{tex} controls texture artifact power with limited effect on class alignment or FID, while s_{sem} drives class conditioning but degrades catastrophically above a threshold. LayerCFG at $(2.0,\ 2.0)$ nearly matches the best CFG FID of 15.49 while improving class alignment. At $(2.0,\ 0.5)$ it reduces NHFP from 1.0835 to 1.0018 at a modest FID cost, a point no single CFG scale can reach.

1 Introduction

Denoising diffusion probabilistic models (DDPMs) (4) have become the standard approach for high-quality image synthesis. Classifier-free guidance (CFG) (5) makes these models class-conditional without requiring a separate classifier: a single network handles both conditional and unconditional denoising, and the guided noise estimate combines their outputs at inference time. The guided estimate at each denoising step is:

$$\tilde{\epsilon}_\theta(x_t, t, y) = \epsilon_\theta(x_t, t, \varnothing) + s\,\underbrace{\big(\epsilon_\theta(x_t, t, y) - \epsilon_\theta(x_t, t, \varnothing)\big)}_{\Delta(x_t,\,t,\,y)}. \tag{1}$$

Increasing s sharpens class conditioning but comes with well-documented failure modes: oversaturation, color bleeding, and high-frequency texture artifacts that grow with s (7). Existing approaches to this problem output rescaling (7) and manifold-constrained inference (1) mitigate the symptoms by modifying how the guided estimate is applied, but they still treat Δ as a single undifferentiated signal.

Whether CFG's failure modes are uniform across spatial frequencies, or concentrated in specific frequency bands of Δ, is not obvious from theory alone. The hierarchical structure of the UNet suggests they might not be. The UNet is hierarchical as its bottleneck and deeper encoder layers process global, low-frequency information that determines class identity and coarse layout, while the shallower decoder layers reconstruct texture, edges, and fine spatial detail. Because Δ is the difference of two full UNet outputs, it carries contributions from all of these levels simultaneously: low-frequency semantic content and high-frequency texture content summed together into a single tensor. CFG then scales this entire mixture by s, with no distinction between the two.

If texture artifacts arise primarily from over-amplifying the high-frequency component of Δ, then the natural fix is to separate the two components and control their amplification independently. This hypothesis is not obviously correct, though: over-amplifying low-frequency content could corrupt semantic structure in ways that manifest as texture-like artifacts, and the two bands may not be as independent as the UNet's layerwise hierarchy suggests. We therefore test it directly.

To investigate, we propose **LayerCFG**, which decomposes Δ into low-frequency (semantic) and high-frequency (texture) bands using a 2-D FFT and applies separate guidance scales s_{sem} and s_{tex} to each. Because standard metrics like FID and IS do not directly measure high-frequency artifact energy, we also introduce NHFP (Normalized High-Frequency Power), which compares the relative spectral energy of generated images to real data. Our central research questions are:

1. Does lowering s_{tex} independently reduce NHFP, and what does that cost in FID and class alignment?
2. Does the 2-D $(s_{\text{sem}}, s_{\text{tex}})$ sweep expose quality/artifact tradeoffs that no single CFG scale can reach?
3. How does FID degrade as s_{sem} grows large, and do the two bands fail independently?

2 Background

Diffusion models. Starting from a clean image $x_0 \sim q(x_0)$, the DDPM forward process (4) adds Gaussian noise over T timesteps:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \tag{2}$$

where $\bar{\alpha}_t = \prod_{i=1}^{t}(1-\beta_i)$ follows a cosine schedule (10) with $T = 1000$. A network $\epsilon_\theta(x_t, t, y)$ is trained to predict ϵ from the noisy image, and clean images are recovered by iteratively denoising from $x_T \sim \mathcal{N}(0, I)$.

Classifier-free guidance. Ho and Salimans (5) handle conditional and unconditional denoising with a single model by randomly dropping the class label y to a null token $\varnothing$ during training at rate p_{drop}. At inference the guided estimate in Eq. 1 is applied. The implied clean-image estimate is:

$$\hat{x}_0 = \frac{x_t - \sqrt{1-\bar{\alpha}_t}\, \tilde{\epsilon}_\theta}{\sqrt{\bar{\alpha}_t}}. \tag{3}$$

When s is large, $\tilde{\epsilon}_\theta$ gets inflated, which pushes $\hat{x}_0$ outside the data range and causes oversaturated samples.

UNet resolution structure. For CIFAR-10 at 32×32, the UNet we use operates at four resolutions: 32×32, 16×16, 8×8, and 4×4 (bottleneck), with channels 64/128/256/512. Each encoder stage is connected to the corresponding decoder stage via skip connections. Class conditioning is applied at every residual block through feature-wise affine modulation (AdaGN), with the null token $\varnothing$ given its own learned embedding.

3 Related Work

Diffusion model foundations. Ho et al. (4) introduced DDPMs and the denoising score-matching objective that underlies most diffusion training today. Nichol and Dhariwal (10) improved sample quality with learned noise schedules and a hybrid training objective. Song et al. (11) unified the diffusion and score-based perspectives through a continuous-time SDE framework.

Guidance and its failure modes. Dhariwal and Nichol (2) showed classifier-guided diffusion could outperform GANs on image benchmarks, making guidance central to conditional generation. CFG (5) eliminated the need for a separately trained classifier. Karras et al. (7) traced quality degradation at large s to instability in network activations and proposed output rescaling as a fix. CFG++ (1) takes a different approach, constraining each inference step to stay geometrically consistent with the data manifold, which reduces oversaturation. Both methods apply a uniform correction across all of Δ. Our approach instead treats its low- and high-frequency components as separate quantities that warrant separate scales.

Hierarchical representations. Deep networks encode coarse structure at their deeper layers and fine spatial detail at shallower ones (12). StyleGAN (6) exploits this explicitly, routing identity, style, and texture through distinct synthesis layers. LayerCFG borrows the same intuition but applies it to the guidance direction at inference time, requiring no changes to the model or its training procedure.

4 Analysis: Frequency Content of the Guidance Direction

The guidance direction $\Delta = \epsilon_c - \epsilon_u$ is a 32×32 tensor shaped like an image. Like any image, it has a spectral decomposition: low-frequency components correspond to global class structure and coarse spatial layout, while high-frequency components carry texture, edges, and fine detail. Standard CFG scales all of this uniformly by s. Our hypothesis is that the high-frequency band is what drives texture artifacts at large s, and that we can reduce those artifacts without hurting class alignment by lowering s_{tex} while holding s_{sem} fixed.

We split Δ into two bands using a 2-D discrete Fourier transform with cutoff $f_{\text{cut}} = 8$ cycles per image (the Nyquist limit for a 32×32 image is 16):

- **Low-frequency** ($|f| \leq f_{\text{cut}}$): global class structure and coarse spatial layout.
- **High-frequency** ($|f| > f_{\text{cut}}$): fine texture, edges, and local patterns.

4.1 Normalized High-Frequency Power (NHFP)

To quantify texture artifact levels in generated images, we define the *Normalized High-Frequency Power* (NHFP). For an image x, let $\hat{x}(f)$ denote the 2-D DFT coefficient at spatial frequency f, and let the per-image high-frequency fraction be:

$$\rho(x) = \frac{\sum_{|f|>f_{\text{cut}}} |\hat{x}(f)|^2}{\sum_{f} |\hat{x}(f)|^2}. \tag{4}$$

NHFP is the ratio of this fraction averaged over generated images to the same quantity over real images:

$$\text{NHFP}(S) = \frac{\mathbb{E}_{x\sim S}\big[\rho(x)\big]}{\mathbb{E}_{x\sim \mathcal{D}}\big[\rho(x)\big]}, \tag{5}$$

where S is the set of generated samples and $\mathcal{D}$ is the CIFAR-10 test set. Note that the hard cutoff at $f_{\text{cut}} = 8$ in ρ is separate from the soft sigmoid mask used inside the guidance step. NHFP $= 1$ means the generated images have the same relative high-frequency energy as real data. Values above 1 indicate excess high-frequency content (texture artifacts), and values below 1 mean the generated images are smoother than real ones. Because we normalize within each image, NHFP is insensitive to overall brightness differences and isolates where in the spectrum the error lies. This makes it a useful complement to FID when studying guidance-induced texture effects, since FID does not distinguish between different types of distributional mismatch.

Algorithm 1 LayerCFG one denoising step

Require: x_t, timestep t, class y, scales $(s_{\text{sem}},\ s_{\text{tex}})$, UNet ϵ_θ, cutoff f_{cut}, sigmoid width τ

$\epsilon_c \leftarrow \epsilon_\theta(x_t, t, y)$
$\epsilon_u \leftarrow \epsilon_\theta(x_t, t, \varnothing)$
$\Delta \leftarrow \epsilon_c - \epsilon_u$
$\hat{\Delta} \leftarrow \mathcal{F}[\Delta]$ ▷ 2-D DFT per channel
$M_{\text{low}}(f) \leftarrow \sigma\big((f_{\text{cut}} - |f|)/\tau\big); \quad M_{\text{high}} \leftarrow 1 - M_{\text{low}}$
$\Delta_{\text{low}} \leftarrow \mathcal{F}^{-1}[\hat{\Delta} \cdot M_{\text{low}}]; \quad \Delta_{\text{high}} \leftarrow \mathcal{F}^{-1}[\hat{\Delta} \cdot M_{\text{high}}]$
return $\epsilon_u + s_{\text{sem}}\,\Delta_{\text{low}} + s_{\text{tex}}\,\Delta_{\text{high}}$

5 Method: LayerCFG

5.1 Frequency-Band Decomposition

We decompose the guidance direction $\Delta = \epsilon_c - \epsilon_u$ into two frequency bands via the 2-D DFT:

$$\Delta_{\text{low}} = \mathcal{F}^{-1}\big[\hat{\Delta} \cdot M_{\text{low}}\big], \quad \Delta_{\text{high}} = \mathcal{F}^{-1}\big[\hat{\Delta} \cdot M_{\text{high}}\big], \tag{6}$$

where $\hat{\Delta} = \mathcal{F}[\Delta]$ is the 2-D DFT of Δ, $M_{\text{low}}(f) = \sigma\big((f_{\text{cut}} - |f|)/\tau\big)$ is a soft sigmoid low-pass mask with transition width $\tau = 1.5$ (we use a soft mask rather than a hard cutoff to avoid Gibbs ringing), and $M_{\text{high}} = 1 - M_{\text{low}}$. The decomposition is applied independently per channel and per sample.

Separate guidance scales are then applied to each band:

$$\tilde{\epsilon}_{\text{LayerCFG}} = \epsilon_u + s_{\text{sem}}\,\Delta_{\text{low}} + s_{\text{tex}}\,\Delta_{\text{high}}. \tag{7}$$

s_{sem} controls the amplification of the semantic (low-frequency) part of Δ, and s_{tex} controls the texture (high-frequency) part. Standard CFG is the special case $s_{\text{sem}} = s_{\text{tex}} = s$.

5.2 Inference Procedure

Algorithm 1 gives the full per-step computation. The two UNet forward passes are identical to standard CFG. The only added cost is the FFT decomposition of Δ, which is negligible compared to the forward passes themselves. No retraining or finetuning is needed.

5.3 Method Overview

Table 1 summarizes the two inference methods we evaluate, and Figure 1 shows the LayerCFG pipeline for one denoising step.

Table 1: Inference methods evaluated. All methods use the same trained UNet.

Method	Guidance	Description
CFG (5)	Scalar	Single scale s applied to full Δ
LayerCFG (ours)	Frequency	Separate s_{sem}, s_{tex} applied to FFT-decomposed Δ_{low}, Δ_{high}

6 Experimental Setup

Dataset and architecture. We use CIFAR-10 (8) (32×32, 10 classes, 50,000 training images). Our model is a class-conditional UNet with channel widths 64/128/256/512 across four resolution levels, following the standard DDPM UNet design. Class conditioning is injected at every residual block via feature-wise affine modulation (AdaGN), and the null token $\varnothing$ gets its own learned embedding. We use a cosine noise schedule (10) with $T = 1000$ and conditioning dropout rate $p_{\text{drop}} = 0.1$.

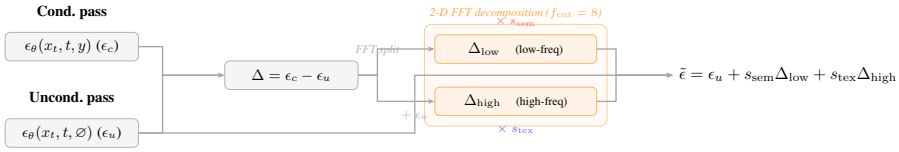

Figure 1: LayerCFG splits $\Delta = \epsilon_c - \epsilon_u$ into a low-frequency (semantic) band and a high-frequency (texture) band using a 2-D FFT with soft sigmoid cutoff at $f_{\text{cut}} = 8$ cycles/image. Separate scales s_{sem} and s_{tex} are applied before recombining with ϵ_u. The two UNet forward passes are identical to standard CFG.

Training. We train a single model with the Adam optimizer (lr $= 2\times10^{-4}$, batch size 128) for 500,000 steps. All inference experiments use this same checkpoint.

Sampling. All samples are generated with DDIM (11) using 100 denoising steps ($\eta = 0$, deterministic).

Evaluation. For each $(s_{\text{sem}},\ s_{\text{tex}})$ configuration we generate 5,000 samples (500 per class) and compute five metrics: (1) **FID** (3), from Inception-v3 pool3 features against 5,000 balanced samples from the CIFAR-10 test set. (2) **IS** (Inception Score, 5 splits). (3) **CA** (classifier alignment): fraction of samples correctly classified by a separately trained ResNet-18, macro-averaged over all 10 classes, trained with AdamW (lr $= 10^{-3}$, weight decay 10^{-4}) for 15 epochs. (4) **NHFP** (Eq. 5), with a hard frequency mask at $f_{\text{cut}} = 8$. (5) **Precision** and **Recall** (9) ($k = 3$ nearest neighbors).

CFG baselines. We evaluate standard CFG at $s \in \{1.0,\ 2.0,\ 3.0,\ 5.0\}$, spanning weak to strong guidance.

LayerCFG sweep. We run LayerCFG over the full grid $s_{\text{sem}} \in \{2.0,\ 3.0,\ 5.0,\ 7.5\} \times s_{\text{tex}} \in \{0.5,\ 1.0,\ 1.5,\ 2.0\}$, giving 16 configurations. The cutoff is fixed at $f_{\text{cut}} = 8$ and $\tau = 1.5$ throughout.

7 Results

7.1 Baseline CFG Behavior

We first characterize how standard CFG behaves across $s \in \{1.0,\ 2.0,\ 3.0,\ 5.0\}$ before introducing the frequency decomposition. Table 2 and Figure 2 report the results.

FID drops from 22.33 at $s = 1.0$ to a best of 15.49 at $s = 2.0$, then rises again as guidance overshoots: 15.98 at $s = 3.0$ and 21.24 at $s = 5.0$. Class alignment improves quickly, from 0.8312 at $s = 1.0$ to 0.9664 at $s = 2.0$, and stays above 0.965 for larger s. IS is nearly flat across the whole sweep (4.784–4.880) and tracks the FID differences poorly.

The clearest signal comes from NHFP. It starts at 1.0892, dips slightly to 1.0835 at $s = 2.0$, then rises sharply: 1.1875 at $s = 3.0$ and 1.4035 at $s = 5.0$. The growth accelerates beyond $s = 2.0$. Precision increases up to $s = 3.0$ (0.6942 $\rightarrow$ 0.7804) as samples become more class-typical, while recall falls steadily (0.5354 $\rightarrow$ 0.4390) the standard fidelity–diversity tradeoff.

The fundamental issue is that no single value of s gives good FID, strong class alignment, and low NHFP at the same time. Getting CA above 0.96 requires $s \geq 2.0$, but at that point NHFP is already elevated and continues to climb.

Table 2: Standard CFG results on CIFAR-10 ($n = 5{,}000$ samples, 100 DDIM steps). NHFP $= 1$ matches real-data texture energy. Values above 1 indicate artifacts.

Method	FID ↓	IS ↑	CA ↑	NHFP	Prec	Rec
CFG $s = 1.0$	22.33	4.784	0.8312	1.0892	0.6942	0.5354
CFG $s = 2.0$	15.49	4.846	0.9664	1.0835	0.7620	0.5116
CFG $s = 3.0$	15.98	4.801	0.9862	1.1875	0.7804	0.4744
CFG $s = 5.0$	21.24	4.880	0.9650	1.4035	0.6950	0.4390

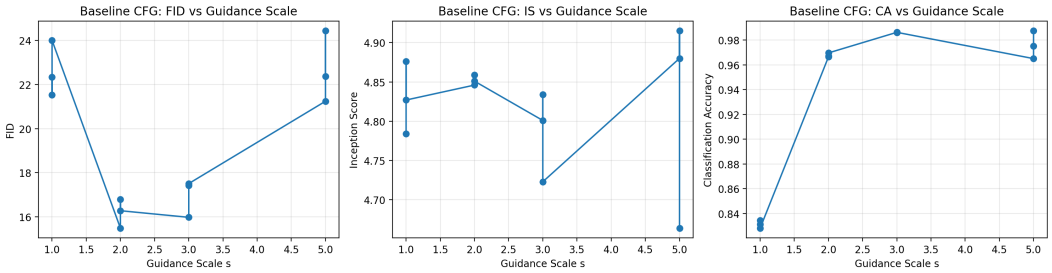

Figure 2: CFG metrics across guidance scales. FID is minimized at $s = 2.0$ while NHFP rises sharply beyond $s = 2.0$, illustrating the quality/artifact tradeoff.

7.2 LayerCFG 2-D Sweep

Table 3 shows results for all 16 configurations, and Figure 3 plots FID as a heatmap over the grid.

Table 3: Full LayerCFG sweep on CIFAR-10 ($n = 5{,}000$ samples). Rows with NHFP < 1 (below real-data texture energy) are marked with †.

s_{sem}	s_{tex}	FID ↓	IS	CA	NHFP	Prec	Rec
2.0	0.5	17.65	4.783	0.9438	1.0018	0.7588	0.5044
2.0	1.0	16.72	4.875	0.9606	1.0141	0.7526	0.5134
2.0	1.5	15.77	4.831	0.9674	1.0473	0.7608	0.5106
2.0	2.0	15.59	4.843	0.9772	1.0967	0.7468	0.5264
3.0	0.5	16.55	4.744	0.9820	0.9828†	0.7848	0.4912
3.0	1.0	16.31	4.698	0.9796	0.9968†	0.7834	0.4676
3.0	1.5	16.36	4.712	0.9794	1.0318	0.7858	0.4748
3.0	2.0	15.88	4.760	0.9856	1.0658	0.7790	0.4900
5.0	0.5	16.64	4.654	0.9792	0.9370†	0.7648	0.4416
5.0	1.0	16.64	4.758	0.9784	0.9517†	0.7522	0.4630
5.0	1.5	16.90	4.839	0.9800	0.9835†	0.7542	0.4640
5.0	2.0	16.99	4.892	0.9838	1.0169	0.7424	0.4510
7.5	0.5	45.80	4.477	0.7994	1.0345	0.5714	0.4290
7.5	1.0	47.15	4.639	0.8090	1.0668	0.5460	0.4250
7.5	1.5	48.11	4.549	0.8122	1.1156	0.5484	0.4196
7.5	2.0	49.72	4.533	0.7992	1.1876	0.5474	0.4088

Three distinct segments emerge from the sweep.

At $s_{\text{sem}} \in \{2.0, 3.0\}$, FID is stable across all s_{tex} values tested (15.59 to 17.65), and s_{tex} acts as a direct control knob on NHFP: lowering it from 2.0 to 0.5 reduces NHFP by roughly 0.09 at a cost of at most 2.1 FID points.

At $s_{\text{sem}} = 5.0$, FID stays stable (16.64–16.99) and CA remains above 0.978, but NHFP behaves unexpectedly: it drops *below* 1.0 when $s_{\text{tex}} \leq 1.5$, falling as low as 0.9370. Strong semantic amplification paired with restrained texture guidance underproduces high-frequency content relative to real images. The samples are class-consistent but smoother than real CIFAR-10, which shows up as declining recall (0.44–0.46).

At $s_{\text{sem}} = 7.5$, FID collapses: 45.8 to 49.7 regardless of s_{tex}. Class alignment falls to around 0.80 and precision to around 0.55, well below the 0.94–0.99 and 0.74–0.79 ranges seen in the three stable rows. Over-amplifying the low-frequency part of Δ alone is enough to destroy image quality even with texture guidance kept low. Notably, NHFP at $s_{\text{sem}} = 7.5$ is only mildly elevated (1.03–1.19) despite the severe FID drop, which tells us the failure here is due to semantic distortion rather than texture artifacts.

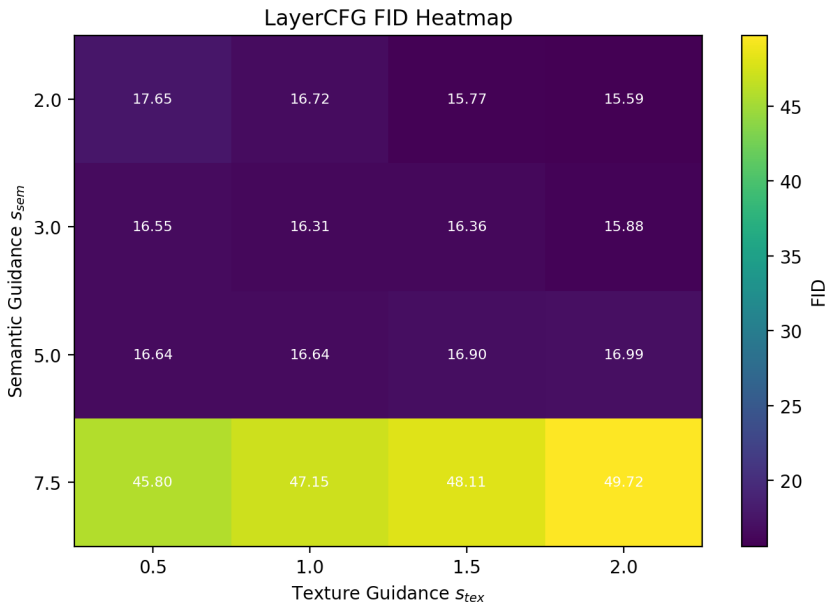

Figure 3: FID heatmap across the $(s_{\text{sem}}, s_{\text{tex}})$ grid. FID is stable at moderate s_{sem} and degrades severely at $s_{\text{sem}} = 7.5$, independent of s_{tex}.

7.3 Best-Method Comparison

Table 4 compares the best CFG configuration against the best LayerCFG results by FID and by NHFP. Figure 4 shows the same comparison visually.

The best-FID LayerCFG point, $(2.0,\ 2.0)$, essentially ties CFG $s = 2.0$ on FID (15.59 vs. 15.49) and IS (4.843 vs. 4.846) while pushing CA from 0.9664 to 0.9772 and picking up 0.0148 in recall. The best-NHFP point, $(2.0,\ 0.5)$, cuts NHFP from 1.0835 to 1.0018 a 7.5% reduction at the cost of raising FID to 17.65. With standard CFG, reaching NHFP near 1.0 would require dropping to $s \approx 1.0$, which gives FID $= 22.33$ (Table 2). LayerCFG reaches a region of (FID, NHFP) space that no single CFG scale can access.

Table 4: Best CFG vs. best LayerCFG configurations.

Method	FID ↓	IS	CA ↑	NHFP	Prec	Rec
CFG $s = 2.0$	15.49	4.846	0.9664	1.0835	0.7620	0.5116
LayerCFG $(2.0,\ 2.0)$ (best FID)	15.59	4.843	0.9772	1.0967	0.7468	0.5264
LayerCFG $(2.0,\ 0.5)$ (best NHFP)	17.65	4.783	0.9438	1.0018	0.7588	0.5044

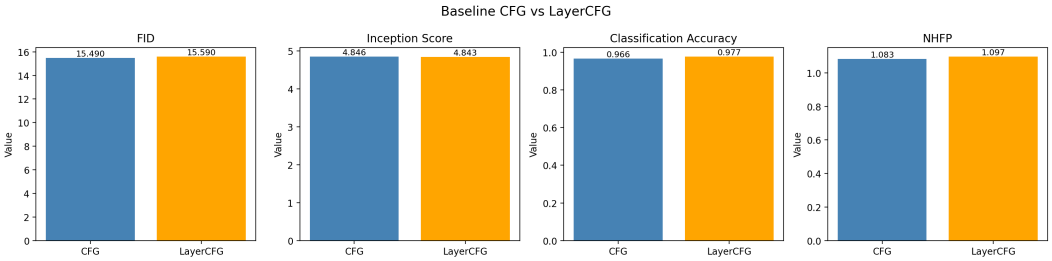

Figure 4: Metric comparison between best CFG and best LayerCFG configurations.

7.4 Qualitative Comparison

Figure 5 is a preliminary qualitative demo using a separately trained 256×256 model, not the CIFAR-10 model used in the quantitative experiments above. We include it to illustrate that the texture softening effect of reducing s_{tex} is visually apparent at higher resolution, where the difference between guidance configurations is easier to see than at 32×32. The 256×256 model was trained on a small subset of data and has not been tuned for sample quality; the comparison is meant to be illustrative, not a quantitative benchmark. CFG $s = 3.0$ (top row) and LayerCFG $(2.0,\ 0.5)$ (bottom row) are shown. The reduction in high-frequency texture energy under LayerCFG is visible in the smoother surface detail, consistent with the NHFP trends measured on CIFAR-10.

Figure 5: Preliminary qualitative demo on a 256×256 model (not the CIFAR-10 model). CFG $s = 3.0$ (top) vs. LayerCFG $(2.0,\ 0.5)$ (bottom). This model has not been tuned for sample quality and is included only to illustrate the visual effect of reducing s_{tex} at higher resolution.

8 Discussion

NHFP control via s_{tex}. The sweep confirms that the high-frequency band of Δ is what drives texture artifact power, at least within the guidance ranges we tested. Reducing s_{tex} from 2.0 to 0.5 while holding $s_{\text{sem}} = 2.0$ fixed drops NHFP from 1.0967 to 1.0018, a real, measurable reduction. The two bands appear to carry largely independent information: the low-frequency band mainly controls class alignment, and the high-frequency band mainly controls texture energy.

Sub-1.0 NHFP at $s_{\text{sem}} = 5.0$. We did not expect NHFP to fall below 1.0. At $s_{\text{sem}} = 5.0$ with $s_{\text{tex}} \leq 1.5$, it drops as low as 0.9370, meaning the generated images have *less* relative high-frequency energy than real CIFAR-10. FID stays reasonable (16.64–16.99) and CA is above 0.978, so from FID alone the samples look fine. But NHFP combined with declining recall (0.44–0.46) indicates the images have become more stereotypically class-like at the cost of textural diversity. This is a case where NHFP catches something that FID misses.

Low-frequency amplification is not "safe". Our expectation was that s_{sem} could be pushed up for stronger class conditioning while keeping texture distortion in check with low s_{tex}. This holds through $s_{\text{sem}} \approx 5.0$, but at $s_{\text{sem}} = 7.5$ FID degrades severely (45–50) and CA drops to around 0.80, regardless of s_{tex}. Over-amplifying the low-frequency band distorts global spatial structure in a way that cannot be recovered by reducing the texture scale.

Connection to hierarchical generation. The semantic/texture decoupling we see here is reminiscent of how StyleGAN (6) separates coarse identity from fine texture control across different synthesis network resolutions. LayerCFG achieves something similar through a frequency split of the guidance direction at inference time, with no changes to the model itself.

Limitations. The cutoff $f_{\text{cut}} = 8$ was chosen by hand and not tuned. A different value might work better, and the right choice likely depends on image resolution. All experiments are on CIFAR-10 at 32×32, so we cannot say whether the same trends hold for larger images or text-conditional models. The FFT-based decomposition is also a fairly coarse tool working directly on UNet intermediate features would give finer-grained control, but would require modifying the model. Finally, 5,000 samples per configuration is likely sufficient for FID on CIFAR-10 but may leave precision/recall estimates noisy.

9 Conclusion

This work started from a simple hypothesis: CFG texture artifacts come mainly from over-amplifying the high-frequency part of the guidance direction Δ. To test it, we built LayerCFG, which decomposes Δ into low-frequency (semantic) and high-frequency (texture) bands via a 2-D FFT and applies separate scales s_{sem} and s_{tex} to each.

On CIFAR-10, LayerCFG at $(s_{\text{sem}},\, s_{\text{tex}}) = (2.0,\, 2.0)$ essentially ties the best CFG baseline on FID (15.59 vs. 15.49) while improving class alignment from 0.9664 to 0.9772. Setting $s_{\text{tex}} = 0.5$ brings NHFP down to 1.0018 at FID 17.65, a combination that no single CFG scale can reach. The most surprising result is that $s_{\text{sem}} = 5.0$ with low s_{tex} drives NHFP below 1.0 while FID stays stable, showing that strong semantic guidance can actually suppress relative texture energy. At $s_{\text{sem}} = 7.5$, FID collapses regardless of s_{tex}, which shows the low-frequency band is not a safe handle to push arbitrarily high.

LayerCFG adds no training cost and minimal inference overhead, and opens up quality/artifact tradeoffs that a single CFG scale cannot reach.

Code

Project code and implementation are available at: `https://github.com/jasmine-le29/ECE285-Final-Project`

References

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